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Probabilistic Palmprint Embedding with Generative Regularization for Open-Set Identification

DAP391m Project

Dự án Trí tuệ Nhân tạo và Khoa học Dữ liệu

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Outline — Nội dung trình bày (10 DS Steps)

Part	Content	Summary
1	Problem & Data Understanding	
2	Feature Engineering & Visualization	
3	Modeling	
4	Evaluation	
5	Application & Conclusion	
6	AI Reflection	

PART 1

Problem & Data Understanding + SQL

Step 1: Problem Understanding	Step 2: Data Understanding	Step 3: EDA	Research Questions	SQL Queries

Step 1: Problem Understanding

- **Palmprint recognition** uses palm lines, texture, and geometry to identify a person.
- Current palmprint systems are usually designed for **closed-set identification**, where every test person already exists in the registered database.
- Key metrics: **Accuracy, FAR, FRR, EER**.
- Main challenges: **illumination changes, hand rotation, scale variation, blur, and cross-dataset differences**.

Therefore, this project focuses on both **closed-set** and **open-set** scenarios:

•Closed-set: identify a known registered person.

•Open-set on the same dataset: reject unseen identities from the same data distribution.

•Open-set on different datasets: reject identities captured under different devices or environments.

Research Questions

- **RQ1:** Why is a new palmprint dataset needed for open-set attendance systems?
- **RQ2:** Can a **Generative Decoder + Contrastive / ArcFace learning** improve palmprint embedding quality?
- **RQ3:** How can the model recognize registered users while rejecting unknown people?
- **RQ4:** If the model is trained on **Camera A** and tested on **Camera B**, can it still reject unknown users?

Step 2: Dataset Understanding

Dataset Name	Image size	Number of Person	Characteristics
IITD	[150, 150] ROI	230	Touchless, indoor, circular illumination, pose and scale variation
Tongji	[128,128] ROI	300	Contactless, controlled device, 12,000 images from 600 palms, 2 sessions
OwnDataset	[128, 128] ROI	231	Real-world campus dataset, 21,977 RGB images, 462 palm classes, varying lighting, pose, and distance

Step 2: Dataset Understanding - IITD

- **IITD** is a public touchless palmprint benchmark collected at IIT Delhi.
- It contains palm images from **230 subjects**, including both left and right hands.
- Each subject has multiple samples captured with **hand pose and scale variations**.
- Images were collected in an **indoor environment** using circular fluorescent illumination.
- The dataset provides original hand images and **cropped ROI palmprint images** for model evaluation.
- Compared with our FPTU-Palm dataset, IITD is more **lab-controlled**, so it is useful for testing cross-dataset domain shift.

Step 2: Dataset Understanding – Tongji

- **Tongji** is a large-scale **contactless palmprint dataset**.
- It contains **12,000 images** from **300 volunteers** and **600 palms**.
- Each palm was captured in **two sessions**, with **10 images per session**.
- The average interval between two sessions was about **61 days**.
- Images were collected using a **dedicated acquisition device**, so the dataset is more **controlled** than real-world campus data.

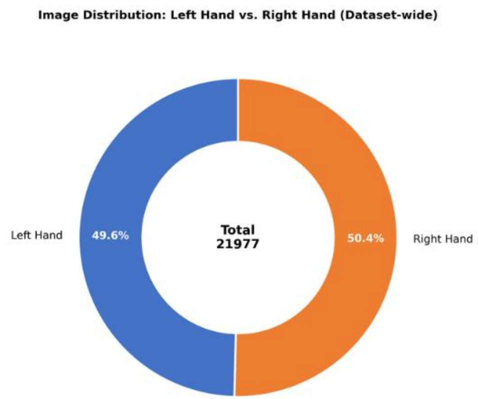
Step 2: Dataset Understanding – Our Collectiton

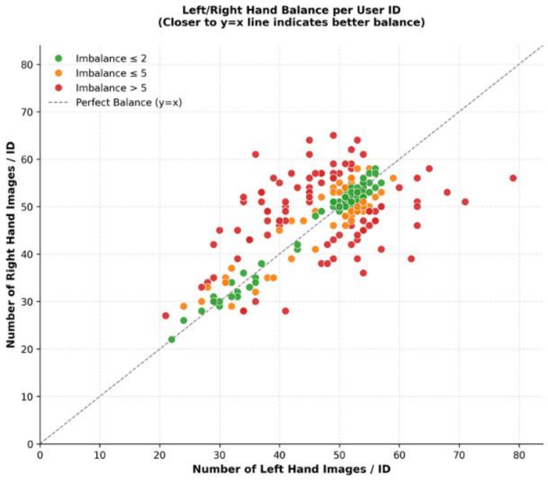
- **FPTU-Palm** is our self-collected palmprint dataset for attendance and biometric authentication.
- It was collected from **231 FPT University students**.
- The dataset contains **21,977 RGB images** from **462 palm classes**: left and right hands.
- Images were captured using a **Razer Kiyo X webcam** at **1920×1080 resolution**.
- Data was collected in a **real-world campus environment** without fixed hand-position constraints.
- Main variations include **lighting, hand pose, camera distance, blur, and intra-class differences**.
- The dataset is useful for studying **open-set recognition** and **cross-dataset domain shift**.

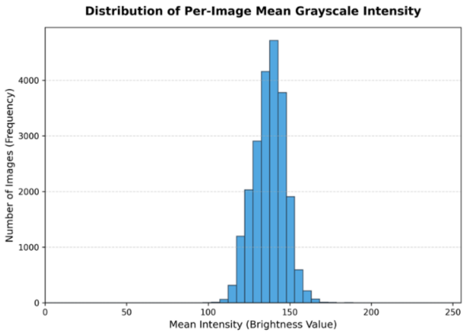
Step 3: EDA Summary

- EDA was conducted on **IITD, Tongji, and FPTU-Palm**.
- We checked **class balance, left/right hand ratio, brightness, sharpness, blur, pose, and scale variation**.
- IITD and Tongji are more **lab-controlled**, while FPTU-Palm is more **real-world and unconstrained**.
- The main data issues are **illumination changes, focus variation, pose variation, and cross-dataset domain shift**.
- These findings justify the use of **augmentation, normalization, robust embedding learning, and open-set evaluation**.

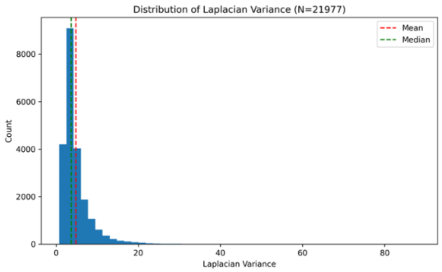
Step 3: EDA Summary



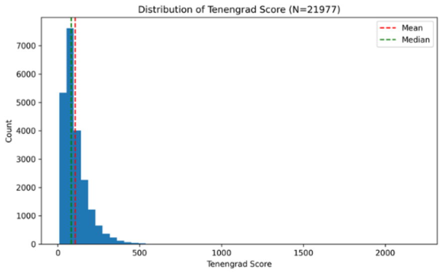




(a) Mean Grayscale Intensity

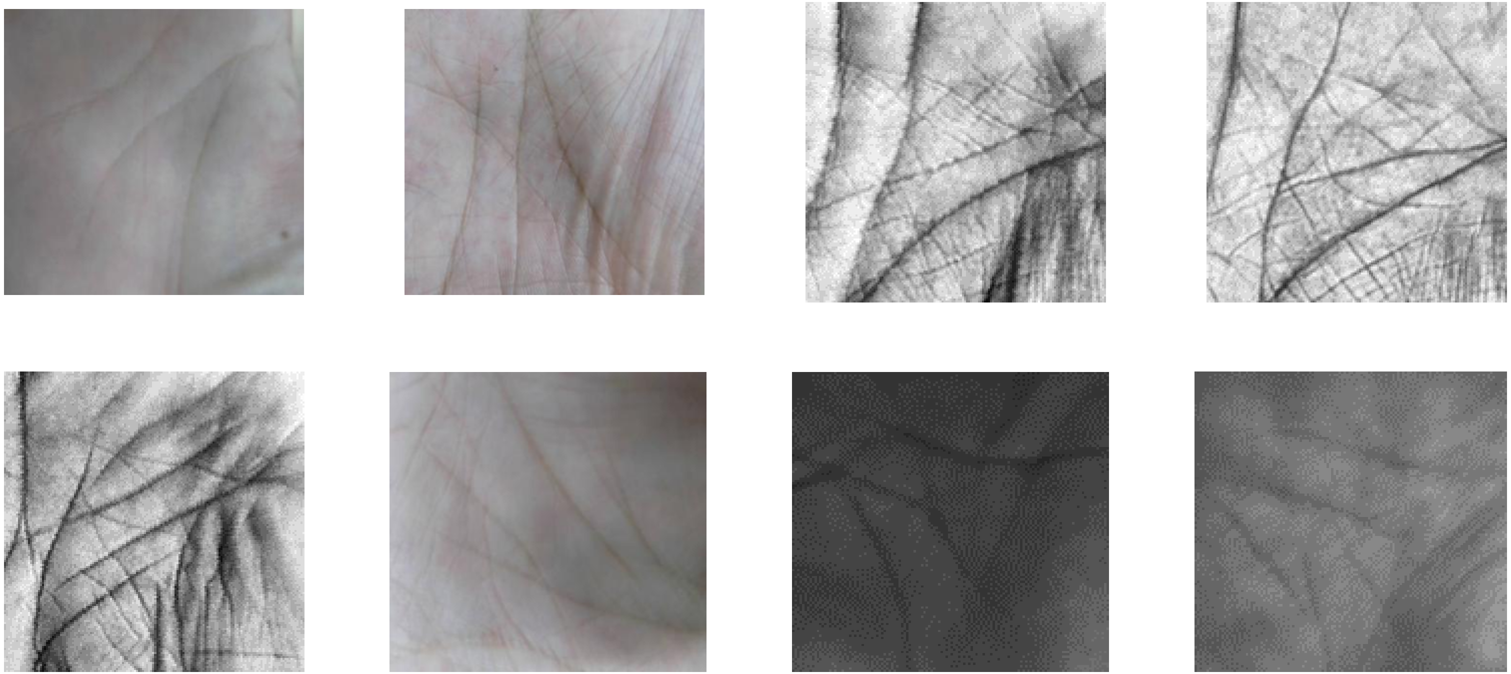


(b) Laplacian Variance



(c) Tenengrad Score

Step 3: EDA Summary – Visualize



Step 4: Feature Engineering

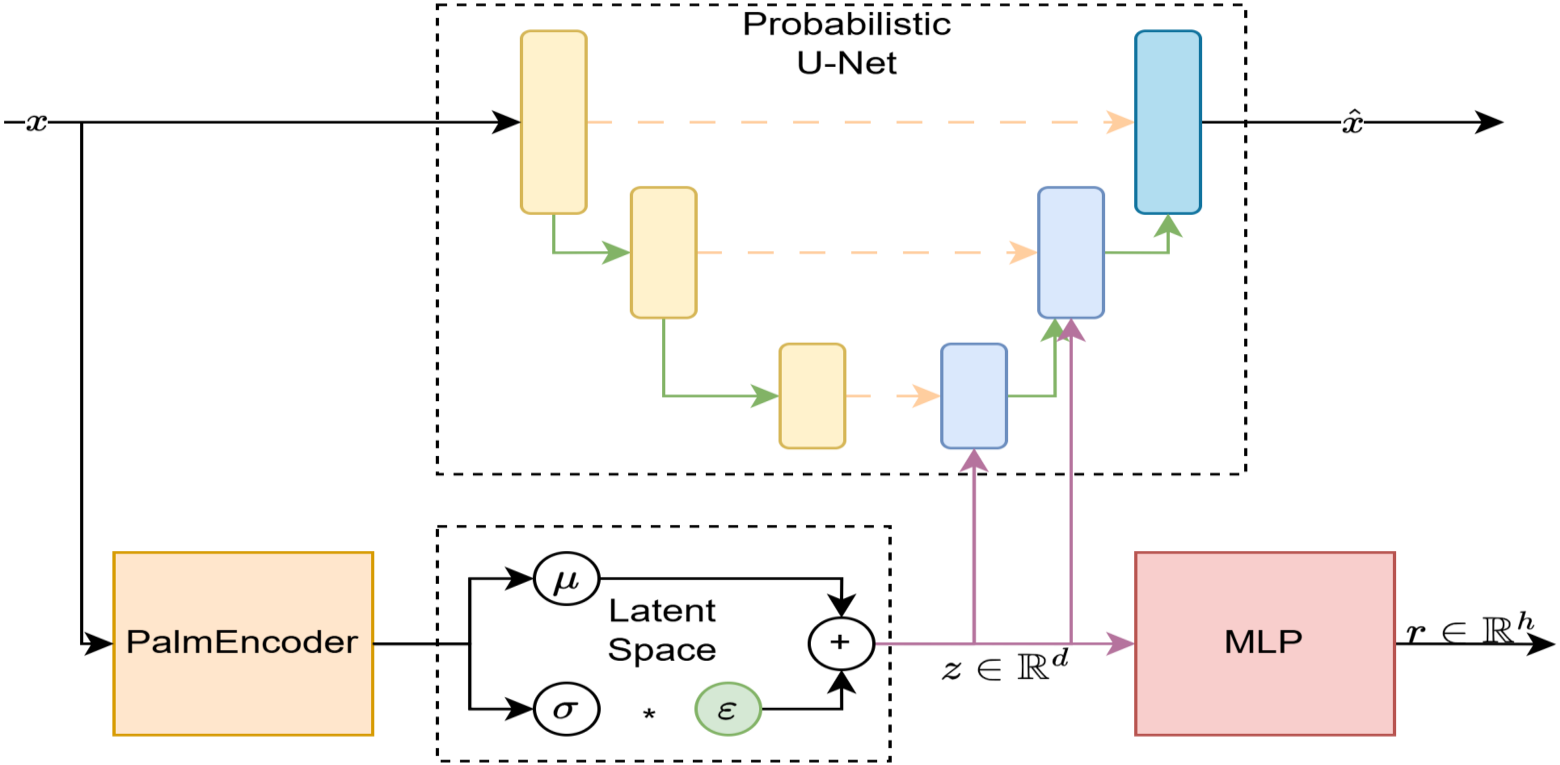
- Each dataset has different image size, color format, and acquisition conditions.
- All images are standardized by **ROI cropping**, **resizing**, and **normalization**.
- Common input size: **256×256** for our training pipeline.
- Data augmentation is applied only during training:
  - Random rotation
  - Random crop / resize crop
  - Color jitter
  - Brightness and contrast adjustment
  - Light blur or noise
- The goal is to reduce the effect of **lighting, pose, scale, and camera-distance variation**.
- Validation and test images are only resized and normalized to ensure fair evaluation.

PART 2

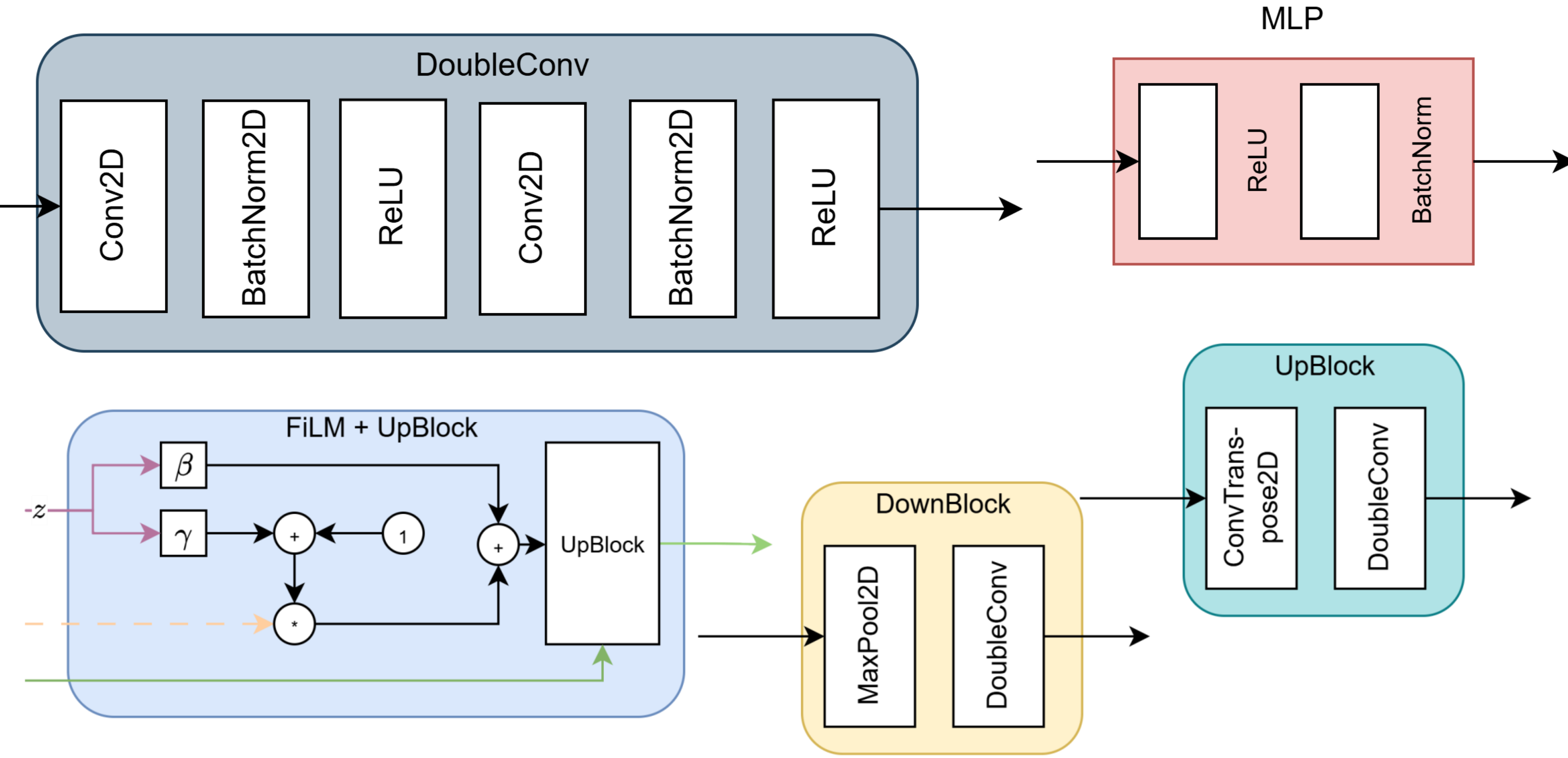
Modeling, Evaluation & AI Application

Step 5: Dataset Partition	Step 6: Modelling	Step 7: Evaluation	Step 8: Tuning	Step 9: Pipeline + App

Overall Pipeline



Overall Pipeline



Training model

Loss function:

$$L = \alpha L_{\text{recon}} + \beta L_{\text{KL}} + \gamma L_{\text{cont}} + \lambda L_{\text{unc}}$$

Reconstruct loss:

$$L_{\text{recon}} = \frac{1}{N} \sum_{i=1}^{N} \|x_i - \hat{x}_i\|^2$$

KL loss:

$$L_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^{J} \left(1 + \log\left(\sigma_j^2\right) - \mu_j^2 - \sigma_j^2\right)$$

ArcFace loss: $L_{\text{cont}} = -\log \dfrac{e^{s \cdot \cos\left(\theta_{y_i} + m\right)}}{e^{s \cdot \cos\left(\theta_{y_i} + m\right)} + \sum_{j=1, j \neq y_i}^{N} e^{s \cdot \cos\left(\theta_j\right)}}$

Uncertainty loss:

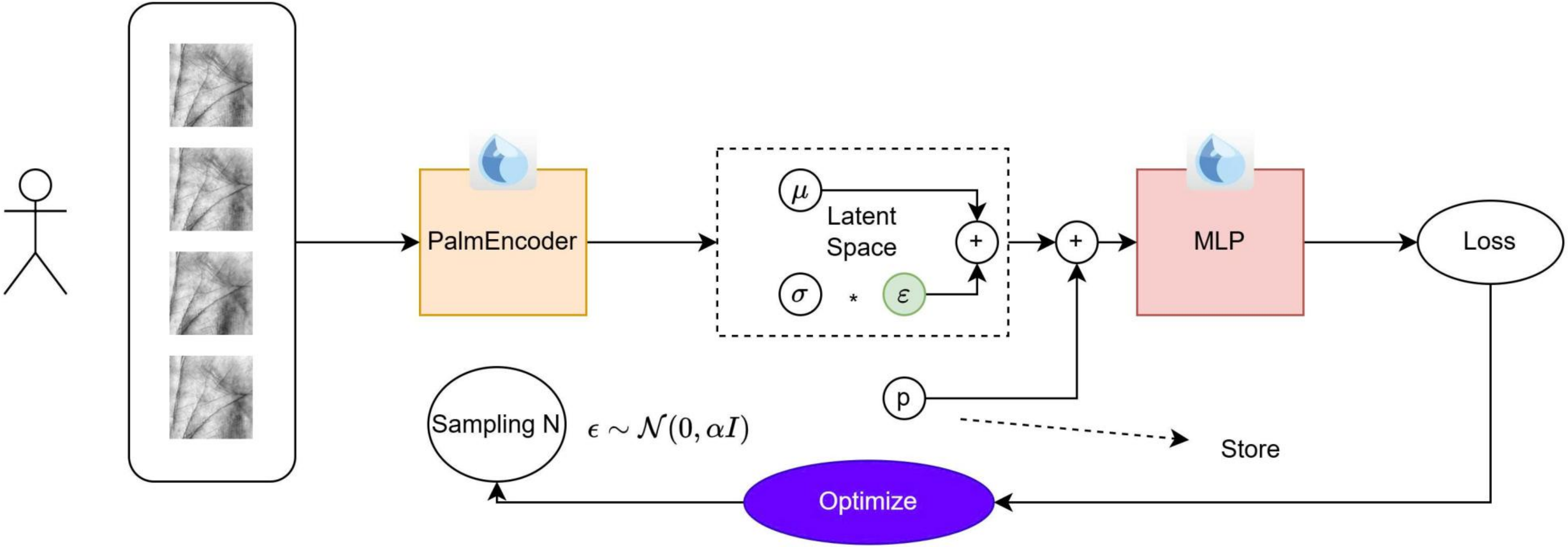
$$L_{\text{unc}} = \frac{1}{2} e^{-s} \|x - \hat{x}\|^2 + \frac{1}{2} s$$

Training strategy

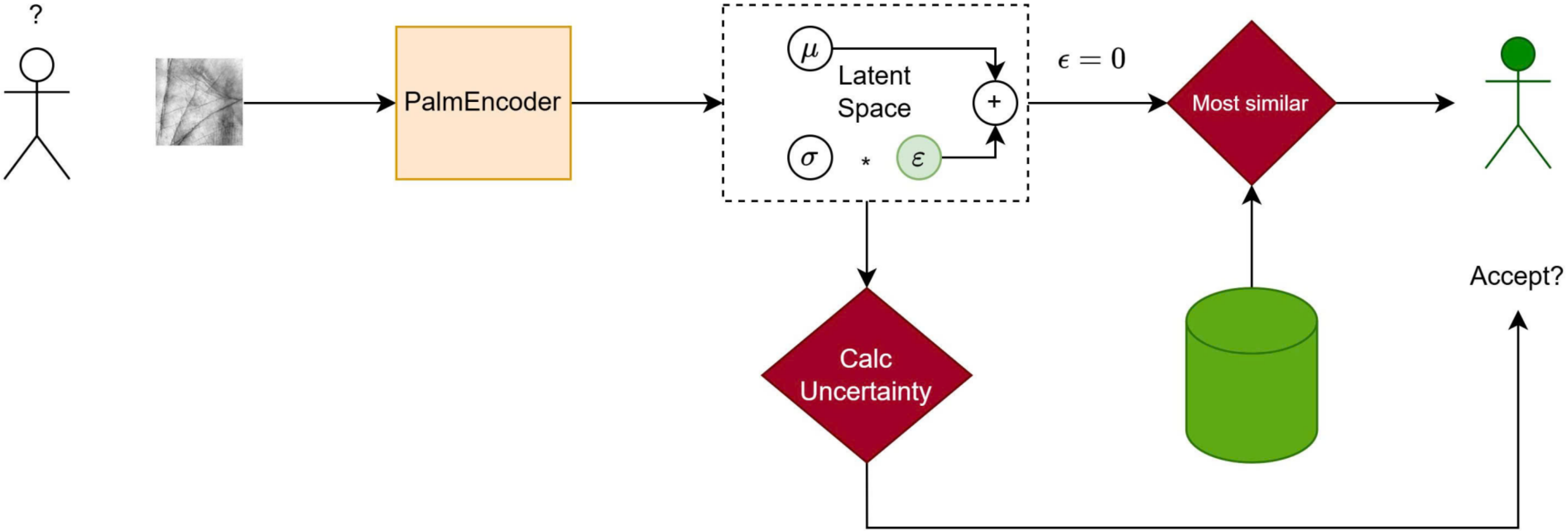
Stage 1 – Stable Identity Learning:
Train the PalmEncoder with contrastive / ArcFace loss to learn discriminative palmprint embeddings.
Stage 2 – Probabilistic Embedding:
After **3,000 steps**, switch from deterministic features to sampled latent vector $\mathbf{z = \mu + \sigma \cdot \varepsilon}$.
Reconstruction Warm-up:
The U-Net decoder is gradually activated from step **2,400 to 7,200** to avoid disturbing early identity learning.
KL Annealing:
The KL weight increases slowly from **0.001 to 0.05**, preventing latent-space collapse.
Uncertainty Regularization:
A constant uncertainty loss is used to make noisy, blurred, or ambiguous samples produce higher uncertainty.
Inference Optimization:
During deployment, the decoder is removed. Only **PalmEncoder + MLP projector** is used for fast attendance verification.

```yaml
switch_to_z_step: 3000


lambda_con:
  type: "constant"
  value: 1.0

lambda_rec:
  type: "linear_step"
  start_step: 2400
  end_step: 7200
  start_value: 0.0
  end_value: 0.15

beta_kl:
  type: "linear_step"
  start_step: 0
  end_step: 10000
  start_value: 0.001
  end_value: 0.05

lambda_unc:
  type: "constant"
  value: 0.1
```

Register new Person



Detect strange person mechanism



Step 2: Dataset Partition

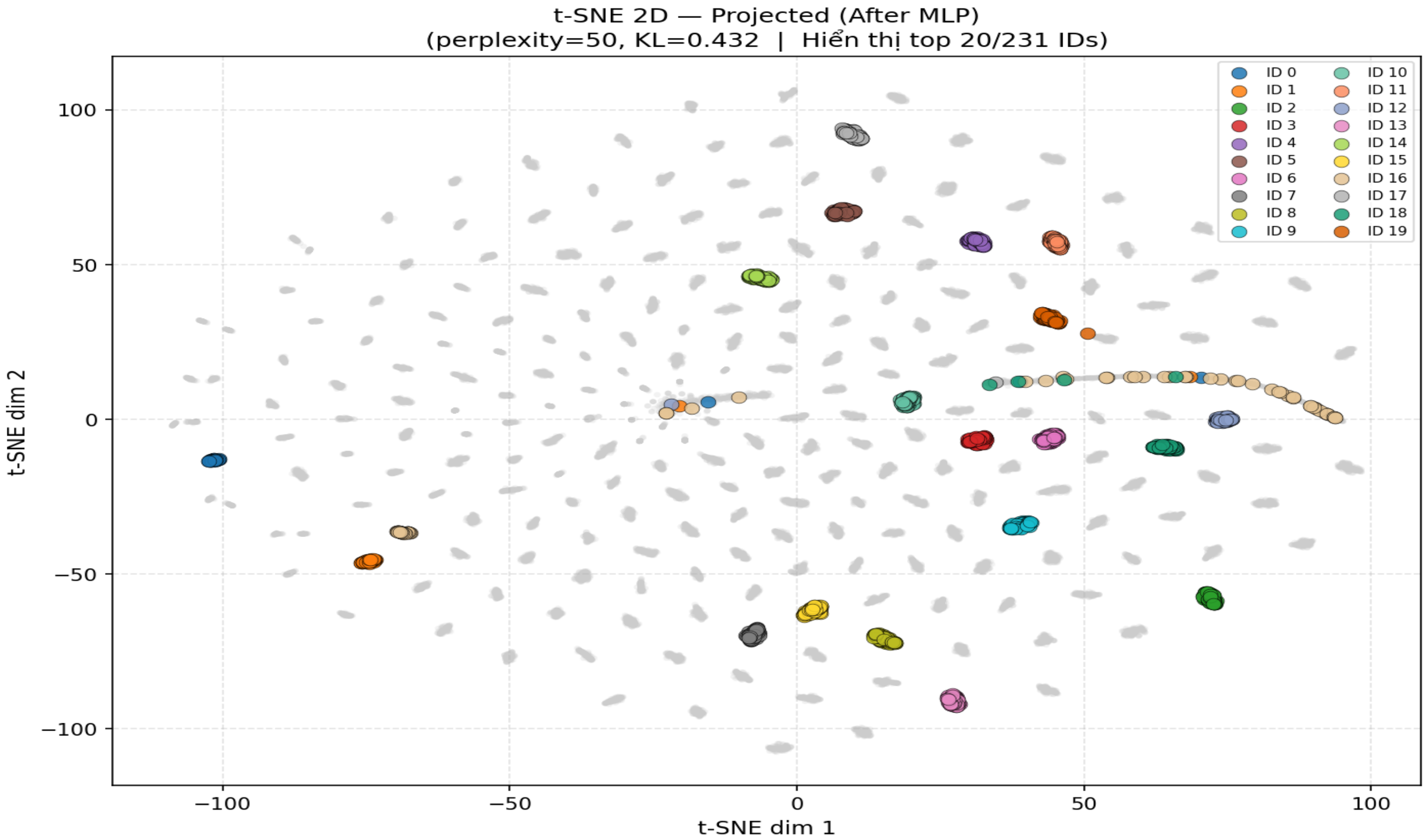
Dataset Name	Train	Register	Strange
IITD	160	40	30
Tongji	200	50	50
OwnOriginal	160	40	31

Step 7 — Data Evaluation

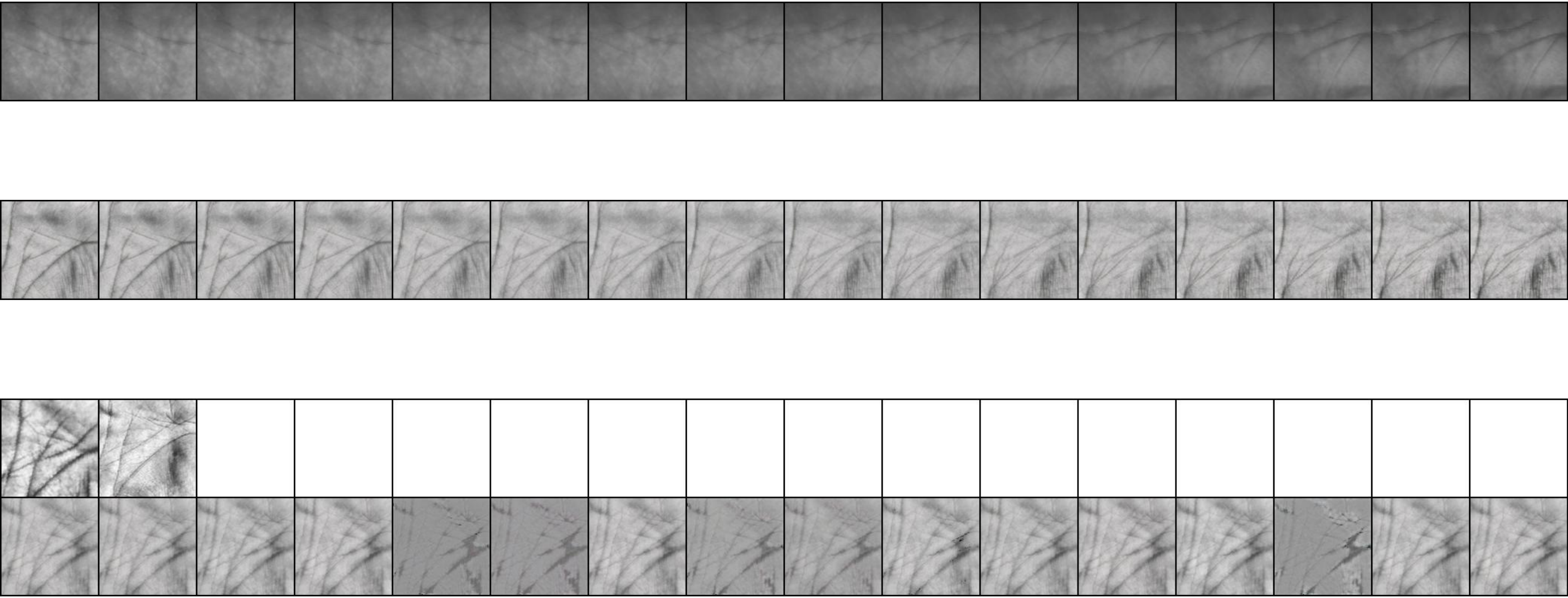
$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

$$EER = FAR(r^*) = FRR(r^*)$$

Step 7 — Data Evaluation (Visualize some results)



Step 7 — Data Evaluation (Visualize some results)



Step 7 — Data Evaluation (Experiment No optimize)

PalmEncoder	IITD		Tongji		OwnDataset	
	ACC	EER	ACC	EER	ACC	EER
VAE (no U-Net)	–	–	–	–	–	–
CCNet	_	_	_	_	_	_
Resnet18	_	_	_	_	_	_
ProPos	_	_	_	_	_	_

Step 7 — Data Evaluation (Experiment Optimize r (Input))

PalmEncoder	IITD		Tongji		OwnDataset	
	ACC	EER	ACC	EER	ACC	EER
VAE (no U-Net)	_	_	_	_	_	_
CCNet	_	_	_	_	_	_
Resnet18	_	_	_	_	_	_
ProPos	_	_	_	_	_	_

Step 7 — Data Evaluation (Experiment Optimize r (Proj))

PalmEncoder	IITD		Tongji		OwnDataset	
	ACC	EER	ACC	EER	ACC	EER
VAE (no U-Net)	–	–	–	–	–	–
CCNet	_	_	_	_	_	_
Resnet18	_	_	_	_	_	_
ProPos	_	_	_	_	_	_